Development of an emissometer for the study of ceramic materials at very high temperatures for energy efficiency processes

Laser heating, calibration and uncertainty of spectral emissivity up to 2000 °C

ABSTRACT

The study of the optical-radiative properties of ceramic materials at very high temperatures (T > 1000 °C) is an area of great technological interest. Understanding these properties allows researchers to evaluate energy efficiency and remotely determine the temperature of materials as diverse as thermal radiators intended to replace flames in industrial electrification processes or high-temperature thermal storage systems.

The project seeks to develop an experimental device that enables the study of the infrared emissivity of ceramic materials at temperatures up to 2000 °C using laser heating (Figure 1), alongside measurement and calibration methods to calculate spectral emissivity and estimate its uncertainty. Once the emissometer is developed, it will be validated using reference materials, such as silicon carbide or alumina, before studying ceramic materials candidates for use as radiators in industrial ovens and thermal storage systems. Finally, the infrared radiation emission mechanisms of the ceramics will be studied using various optical models to interpret the results and determine which materials possess the appropriate properties for these technologies.

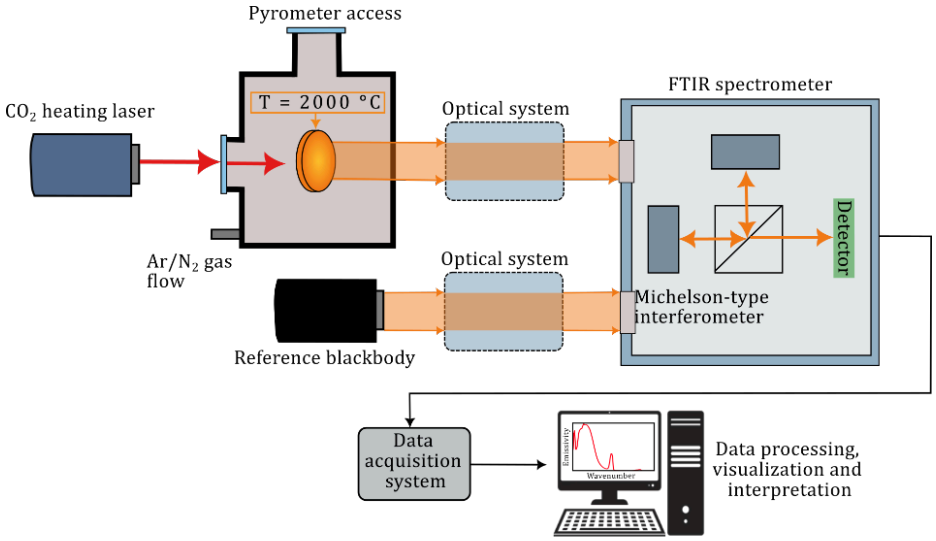


Figure 1. Proposed experimental configuration for spectral emissivity measurements up to 2000 °C

REQUIREMENTS

Nobody is expected to bring all of this. The degrees are the formal part; the rest is welcome and is taught along the way.

- Bachelor's degree in Physics, Electronic Engineering, Industrial Engineering, the Dual Degree in Physics and Electronic Engineering or a related field, and a Master's degree in Materials Science, Electronic Engineering, Mechanical Engineering, Industrial Engineering or a related area, as required for admission to a doctoral programme (or expected completion before the start date).
- Welcome, not required: some background in instrumentation, automatic control or experimental physics, and in infrared optics or measurement uncertainty; some programming, ideally in Python.
- Interest in building, calibrating and validating an experimental device, willingness to work in a team, and English good enough to read, write and present scientific work (B2 or higher).

Full or partial funding is available for the thesis projects.

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